# EXP 5: COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS AND EXPLAIN WITH VARIOUS TEST SCENARIOS

**1. Define the Two Prompt Types**

**1. Naïve Prompt**

A **naïve prompt** is broad, vague, or unstructured. It gives minimal instruction and relies on the AI to guess the context.

Example:
“Tell me about blockchain.”

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**2. Basic Prompt**

A **basic prompt** is clearer, structured, and gives more context or instructions to guide the AI.

Example:
“Explain blockchain technology in simple terms for undergraduate students. Include definition, working principle, and two real-world applications.”

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**2. Test Scenarios**

We tested four scenarios:

1. Creative Story Generation
2. Answering a Factual Question
3. Summarizing a Concept
4. Providing Advice

For each scenario we used **both a naïve prompt and a basic prompt**.

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**Scenario 1 – Creative Story**

**Naïve Prompt**

Write a story about a robot.

**Output**

A robot named Zeta lived in a small laboratory. One day, it became curious about the outside world. It left the lab and explored the city, helping people fix machines and solve problems. Eventually, the robot became known as the helpful robot of the city and people appreciated its intelligence and kindness.

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**Basic Prompt**

Write a **150-word creative story about a friendly robot exploring Earth for the first time. Include a beginning, conflict, and resolution. Target audience: school students.**

**Output**

Riko was a friendly robot sent from a distant planet to explore Earth. When he landed in a busy city park, everything amazed him—trees, birds, and laughing children.

One day, Riko noticed a small drone stuck in a tree. The children were upset because it was their science project. Riko tried climbing but slipped because his metal hands were not designed for trees.

Thinking quickly, he used his sensors to calculate the perfect angle and gently shook the branch using a nearby stick. The drone fell safely onto the grass.

The children cheered and thanked Riko. From that day on, Riko helped people solve problems using his intelligence and kindness. Earth quickly became his favorite place in the universe.

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**Scenario 2 – Factual Question**

**Naïve Prompt**

What is blockchain?

**Output**

Blockchain is a digital ledger technology used to store transactions across multiple computers securely. It is decentralized and commonly used in cryptocurrencies like Bitcoin.

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**Basic Prompt**

Explain **blockchain technology** in simple terms. Include
• Definition
• How it works
• Two real-world applications

**Output**

Blockchain is a **decentralized digital ledger** that records transactions across many computers. Instead of storing data in one central database, blockchain distributes the information across a network.

**How it works**

1. A transaction is requested.
2. The transaction is grouped into a block.
3. The block is verified by network nodes.
4. The verified block is added to the chain of previous blocks.

**Real-world applications**

• **Cryptocurrency** – Used in Bitcoin and Ethereum for secure financial transactions.
• **Supply Chain Management** – Tracks goods from production to delivery to ensure transparency.

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**Scenario 3 – Summarizing a Concept**

**Naïve Prompt**

Summarize artificial intelligence.

**Output**

Artificial intelligence is a technology that allows machines to perform tasks that normally require human intelligence such as learning, reasoning, and problem solving.

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**Basic Prompt**

Summarize **Artificial Intelligence (AI)** in **5–6 sentences** explaining its definition, main types, and common applications.

**Output**

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines capable of performing tasks that normally require human intelligence. These tasks include learning, decision-making, and problem solving. AI systems

work using algorithms, large datasets, and machine learning techniques. There are different types of AI such as **narrow AI**, which performs specific tasks, and **general AI**, which aims to perform any intellectual task a human can do. AI is widely used in applications like virtual assistants, recommendation systems, healthcare diagnosis, and self-driving cars. As technology advances, AI continues to transform many industries.

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**Scenario 4 – Advice / Recommendation**

**Naïve Prompt**

How can I study better?

**Output**

To study better you should manage your time, avoid distractions, and review your notes regularly.

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**Basic Prompt**

Give **5 practical study tips for college students preparing for exams. Include time management, concentration techniques, and revision strategies.**

**Output**

1. **Create a Study Schedule** – Plan daily study sessions and allocate time for each subject.
2. **Use Active Learning** – Practice problems, summarize notes, and teach concepts to others.
3. **Avoid Distractions** – Study in a quiet place and limit phone usage during study time.
4. **Use Revision Techniques** – Review notes regularly and use flashcards or mind maps.
5. **Take Breaks** – Follow techniques like the Pomodoro method (25 minutes study + 5 minutes break) to maintain focus.

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### 3. Comparison Table

| Scenario | Prompt Type | Quality | Accuracy | Depth |
|---|---|---|---|---|
| Creative Story | Naïve | Simple story | Accurate | Low detail |
| Creative Story | Basic | Structured and engaging | Accurate | More detailed |
| Factual Question | Naïve | Short explanation | Accurate | Limited |
| Factual Question | Basic | Clear explanation | Highly accurate | Detailed |
| Summarization | Naïve | Very short | Accurate | Minimal |
| Summarization | Basic | Well structured | Accurate | Moderate |
| Advice | Naïve | General tips | Accurate | Limited |
| Advice | Basic | Organized tips | Accurate | More informative |

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### 4. Analysis

**Quality**

Basic prompts produced **more structured and readable responses**. The outputs often included lists, explanations, and examples.

**Accuracy**

Both naïve and basic prompts produced **accurate information**, but basic prompts improved clarity.

**Depth**

Basic prompts generated **more detailed responses**, while naïve prompts produced shorter answers.

**Observations**

- Naïve prompts work for **simple questions**.
- Basic prompts are better for **complex tasks** like storytelling, explanations, and recommendations.

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**5. Summary of Findings**

The experiment shows that **prompt clarity significantly impacts AI output quality**. Basic prompts guide the AI to produce more structured, informative, and detailed responses.

**Key Insights**

- Clear instructions improve response quality.
- Adding context increases depth.
- Structured prompts produce more useful results.

Therefore, when using ChatGPT, it is recommended to use **basic or refined prompts with clear instructions and context** to obtain optimal results.